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On generalized multiplicative cascades

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Abstract

We consider a generalized Mandelbrot's martingale $\{Y_n\}$ and the associated Mandelbrot's measure μ_ω on marked trees. If the limit variable $Z = \lim Y_n$ is not degenerate, we study the asymptotic behavior at infinity of its distribution; in the contrary case, we prove that there is an associated natural martingale Y_n^* converging to a non-negative random variable Z^* with infinite mean. Both Z and Z^* lead to non-trivial solution of a distributional equation which extends the notion of stable laws. Precise results are obtained about Hausdorff measures and packing measures of the support of the Mandelbrot's measure.

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0. Introduction

We consider a non-negative martingale, $\{Y_n\}_{(n\geqslant 1)}$, defined as sums of products of random weights indexed by nodes of a Galton–Watson tree. There is a natural random measure, μ_ω, defined on the boundary of the Galton–Watson tree, with mass $||\mu_\omega|| = Z(\omega) := \lim_{n\to\infty} Y_n(\omega)$. The classical model of Mandelbrot (1974) corresponds to the case where the tree is a fixed r-ary tree ($r\geqslant 2$ being a constant) and all the weights are independent and identically distributed; this classical model and its variations have been studied by many authors since the work of Kahane and Peyrière (1976), see for example Guivarc'h (1990), Collet and Koukiou (1992), Franchi (1993), Liu (1993), Waymire and Williams (1995), Chauvin and Rouault (1996), Molchan (1996) and Barral (1997). The advantage of the general model presented here is that it unifies the study of cascades and branching random walks, as well as some other subjects. Although our model is much more general (and is therefore applicable to many fields), we shall see that all the classical results of Kahane and Peyrière (1976) and Guivarc'h

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(1992) still hold, and we sometimes obtain some results which are also new even for the canonical model.

Let us explain briefly what we obtain in this paper. If the limit variable Z is non-degenerate, we give a necessary and sufficient condition for existence of moments of any given order $p > 1$ (Theorem 2.1), and prove that under some moment conditions, there is a number $\chi > 1$ (explicitly determined) such that the limit of $x^{\chi}P(Z > x)$ exists and is strictly positive and finite as $x \to \infty$ (Theorem 2.2); we also give a necessary and sufficient condition for existence of exponential moments of some order $s > 0$ (Theorem 2.3), and obtain precise results on Hausdorff measures of the support of the measure μ_ω (Theorem 2.4). If Z is degenerate, we prove that, instead of $\{Y_n\}$, there is another natural normalized martingale, $\{Y_n^*\}$, converging to a non-degenerate random variable $Z^* \geqslant 0$ (Theorem 2.5). Both Z and Z^* lead to solutions (with or without mean) of the distributional equation $Z = \sum_{i=1}^{\infty} A_i Z_i$, where $(A_1, A_2, \ldots)$ is a given random variable with values in $[0,\infty) \times [0,\infty) \times \cdots$ and independent of $(Z_1, Z_2, \ldots)$, the Z_i's being independent copies of Z.

The model is carefully described in Section 1, where a series of examples is given. The main results are stated in Section 2. Some preliminary results about the random difference equation are stated in Section 3. The proofs of main theorems are given in Sections 4–6.

1. The model, and some examples

Let $\mathbb{N}^\star = \{1, 2, \cdots\}$ be the set of positive integers with the discrete topology. Put $\mathbb{N} = \{0\} \cup \mathbb{N}^\star$ and write

$$\boldsymbol{U} = \bigcup_{k=0}^{\infty} (\mathbb{N}^\star)^k$$

for the set of all finite sequences $\boldsymbol{i} = i_1 i_2 \ldots i_n$ ($i_k \in \mathbb{N}^\star$), where by convention $\mathbb{N}^{\star 0} = \{\emptyset\}$ contains the null sequence $\emptyset$. Let

$$\boldsymbol{I} = (\mathbb{N}^\star)^{\mathbb{N}^\star}$$

be the set of all infinite sequences $\boldsymbol{i} = i_1 i_2 \ldots$ ($i_k \in \mathbb{N}^\star$) with the product topology. If $\boldsymbol{i} = i_1 i_2 \ldots i_n$ ($n \leqslant \infty$) is a sequence, we write $|\boldsymbol{i}| = n$ for its length, and $\boldsymbol{i}|k = i_1 i_2 \ldots i_k$ ($k \leqslant n$) for the curtailment of $\boldsymbol{i}$ after k terms; conventionally, $|\emptyset| = 0$ and $\boldsymbol{i}|0 = \emptyset$. If $\boldsymbol{i} \in \boldsymbol{U}$ and $\boldsymbol{j} \in \boldsymbol{U}$ or $\boldsymbol{I}$ we write $\boldsymbol{ij} = (\boldsymbol{i}, \boldsymbol{j})$ for the sequence obtained by juxtaposition. In particular $\boldsymbol{i}\emptyset = \emptyset\boldsymbol{i} = \boldsymbol{i}$. We partially order $\boldsymbol{U}$ by writing $\boldsymbol{i} \leqslant \boldsymbol{j}$ to mean that for some $\boldsymbol{i}' \in \boldsymbol{U}$, $\boldsymbol{j} = \boldsymbol{ii}'$, and we use a similar notation if $\boldsymbol{i} \in \boldsymbol{U}$ and $\boldsymbol{j} \in \boldsymbol{I}$. If $\boldsymbol{i}$ and $\boldsymbol{j}$ are two sequences, we write $\boldsymbol{i} \wedge \boldsymbol{j}$ for the maximal common sequence of $\boldsymbol{i}$ and $\boldsymbol{j}$, that is, the maximal sequence $\boldsymbol{q}$ such that $\boldsymbol{q} \leqslant \boldsymbol{i}$ and $\boldsymbol{q} \leqslant \boldsymbol{j}$.

A tree T is a subset of $\boldsymbol{U}$ satisfying three conditions (cf. Neveu, 1986): (i) $\emptyset \in T$; (ii) if $\boldsymbol{ij} \in T$, then $\boldsymbol{i} \in T$; (iii) if $\boldsymbol{i} \in T$ and $j \in \mathbb{N}^\star$, then $\boldsymbol{i}j \in T$ if and only if $1 \leqslant j \leqslant N_{\boldsymbol{i}}$ for a positive integer $N_{\boldsymbol{i}}$. We shall write N for $N_\emptyset$. The boundary of a tree T is defined as

$$\partial T = \{\boldsymbol{i} \in \boldsymbol{I}\colon \boldsymbol{i}|n \in T \text{ for all } n \in \mathbb{N}\}.$$

As a subset of $\boldsymbol{I}$, it is a metrical and compact topological space with

$$B(\boldsymbol{i}) = \{\boldsymbol{j} \in \partial T: \boldsymbol{i} \leqslant \boldsymbol{j}\}, \quad \boldsymbol{i} \in T,$$

its topological basis; a possible choice of metric is

$$d_c(\boldsymbol{i}, \boldsymbol{j}) = c^{-|\boldsymbol{i} \wedge \boldsymbol{j}|},$$

where c is a given number in $(1, \infty)$. The set $B(\boldsymbol{i})$ is then a ball of radius $c^{-|\boldsymbol{i}|}$.

An element of T is called a node. Each node $u \in T$ is marked with a vector $\eta_u = (A_{u1}, A_{u2}, \ldots)$ of $\mathbb{R}_+^{\mathbb{N}^\star}$, where $\mathbb{R}_+ = [0, \infty)$. If $1 \leqslant j \leqslant N_u$, we can imagine that the number A_{uj} is associated with the edge (u, uj) linking the nodes u and uj; the values A_{uj} for $j > N_u$ are of no influence for our purpose, and will be taken as 0 for convenience. The marked tree will be denoted by $(T, (\eta_u, u \in T))$.

Let $\mathbb{T}$ be the set of all trees, and Ω be the set of all marked trees ω (marked as above). An element ω of Ω will be written as $(T(\omega), (\eta_u, u \in T(\omega)))$, where $T(\omega)$ is the underlying tree. We may regard T as the canonical projection from Ω to $\mathbb{T}$. Thus T may stand for a tree or an operator, according to the context. If ω is a marked tree and if $\boldsymbol{i} \in T(\omega)$, we write $T_{\boldsymbol{i}}(\omega) = \{\boldsymbol{j} \in \boldsymbol{U}: \boldsymbol{ij} \in T(\omega)\}$ for the shifted tree of $T(\omega)$ at $\boldsymbol{i}$. Note that $T_{\boldsymbol{i}}(\omega) \in \mathbb{T}$. Denote by

$$z_k(\omega) = \{\boldsymbol{i} \in T(\omega): |\boldsymbol{i}| = k\}$$

the set of nodes of $T(\omega)$ with length k ($k \in \mathbb{N}$), and consider the filtration

$$\mathscr{F}_k = \sigma\{(N_{\boldsymbol{i}}, A_{\boldsymbol{i}1}, A_{\boldsymbol{i}2}, \ldots): |\boldsymbol{i}| < k, \boldsymbol{i} \in T(\omega)\}, \quad k \geqslant 1.$$

Let $\mathscr{F} := \sigma(\mathscr{F}_k, k \geqslant 1)$. For simplicity, we write $(N, A_1, A_2, \ldots)$ for $(N_\emptyset(\omega), A_{\emptyset 1}(\omega), A_{\emptyset 2}(\omega), \ldots)$.

By a result of Neveu (1986), for each probability distribution q on $\mathbb{N} \times \mathbb{R}_+^{\mathbb{N}^\star}$, there is a probability law $P = P_q$ on $(\Omega, \mathscr{F})$ such that

(i) the distribution of the random variable $(N, A_1, A_2, \ldots)$ is q;

(ii) given $\mathscr{F}_k$, the random variables $(N_{\boldsymbol{i}}, A_{\boldsymbol{i}1}, A_{\boldsymbol{i}2} \ldots)$, $\boldsymbol{i} \in z_k(\omega)$, are conditionally independent, and their conditional distribution is q.

The property (ii) is referred as the *branching property*.

Assume that the initial distribution is normalized such that

$$E\left(\sum_{i=1}^{N} A_i\right) = 1$$

(in this paper we always make the convention that any sum over an empty set is taken to be 0). If $u \notin T(\omega)$, the values $A_u(\omega)$ are of no influence for our problem, and may be non-defined; however, for convenience, we set

$$A_u = 0 \quad \text{if } u \in \boldsymbol{U} \setminus T(\omega).$$

In particular, for all $u \in T(\omega), A_{ui} = 0$ if $u > N_u$. Put

$$X_\emptyset = 1, \; X_u = A_{u_1} A_{u_1 u_2} \cdots A_{u_1 \ldots u_n} \quad \text{if } u = u_1 \ldots u_n \in \boldsymbol{u}$$

and

$$Y_n = \sum_{|v| = n, v \in T(\omega)} X_v, \quad n \geqslant 1.$$

Then $(Y_n, \mathscr{F}_n)$ $(n \geqslant 1)$ is a (non-negative) martingale, so the limit $Z = \lim_{n\to} Y_n$ exists almost surely (a.s.) with $EZ \leqslant 1$. Similarly, we put

$$Z_u = \lim_{n\to\infty} \sum_{v=v_1\dots v_n \in T_u(\omega)} A_{uv_1} A_{uv_1v_2} \cdots A_{uv_1\dots v_n} \quad \text{if } u \in T(\omega),$$

and $Z_u = 1$ if $u \in \boldsymbol{U} \setminus T(\omega)$. Then $Z = Z_\emptyset$, and, by the branching property, given $\mathscr{F}_n$, the random variables $Z_u, u \in z_n(\omega)$, are conditionally independent, and their conditional law is the distribution of Z. It is easily seen that for almost all $\omega \in \Omega$ and all $u \in T(\omega)$,

$$X_u Z_u = \sum_{i=1}^{N_u} X_{ui} Z_{ui}.$$

Therefore for almost all $\omega \in \Omega$, there is a unique Borel measure, called μ_ω, defined on $\partial T(\omega)$, such that

$$\mu_\omega(B(u)) = X_u Z_u \quad \text{for all } u \in T(\omega).$$

We extend it as a Borel measure on $\boldsymbol{I}$ by letting $\mu_\omega(A) = \mu_\omega(A \cap \partial T(\omega))$. Then μ_ω is a random measure on $\boldsymbol{I}$ with mass $Z(\omega)$. This measure was introduced in Liu (1993) for calculations of exact Hausdorff dimensions of some random fractals. A similar measure in an Euclidean space $\mathbb{R}^d$ was constructed in Mauldin and Williams (1986). One can easily check that for all $u \in T(\omega)$,

$$\mu_\omega(B(u)) = Z(\omega) \lim_{n\to\infty} \frac{\sum_{|v|=n, v\in T(\omega), u\leqslant v} X_v}{\sum_{|v|=n, v\in T(\omega)} X_v}.$$

The preceding identity on Z_u shows, in particular, that the distribution of Z satisfies the equation

$$Z = \sum_{i=1}^{N} A_i Z_i, \tag{E}$$

where given $(N, A_1, A_2, \dots)$, the random variables Z_i $(1 \leqslant i \leqslant N)$ are conditionally independent, and their conditional distribution is the law of Z. In terms of characteristic functions or Laplace transforms, it reads

$$\phi(t) = E \prod_{i=1}^{N} \phi(A_i t), \tag{E$'$}$$

where $\phi(t) = E(\mathrm{e}^{\mathrm{i}tZ})$ or $E(\mathrm{e}^{-tZ}), t \in \mathbb{R}$ or $\mathbb{R}_+$, and the empty product is taken to be 1.

Let us now give some typical examples to illustrate the model. The results that we shall obtain can be applied to each of these examples.

Example 1.1 (Self-similar cascades on a c-ary tree). $N = c$ is a constant $\geqslant 2$, and $A_1, \dots, A_c$ are independent and identically distributed (i.i.d.). In this case,

$$T(\omega) = T := \bigcup_{k=0}^{\infty} \{1, \dots, c\}^k \quad (\{1, \dots, c\}^0 = \{\emptyset\})$$

is a fixed c-ary tree,

$$\partial T(\omega) = \partial T = \{1, \dots, c\}^{\mathbb{N}^\star}$$

is the set of infinite sequences of integers between 1 and c, $A_u = W_u/c$ for all $u \in T$, where $\{W_u\}_{(u\in T)}$ is a family of non-negative i.i.d. random variables with $EW_1 = 1$. Let λ be the natural Borel measure on ∂T such that

$$\lambda(B(u)) = c^{-n} \quad \text{for all } u = u_1 \ldots u_n \in T,\ n > 0,$$

and let $\{\mu_\omega^{(n)}\}_{(n>0)}$ be a sequence of (random) Borel measures on ∂T such that, on each $B(u)$ with $u = u_1 \ldots u_n \in T$,

$$\mu_\omega^{(n)} \text{ has a density } A_{u_1} \ldots A_{u_1 \ldots u_n} c^n$$

with respect to λ. The Mandelbrot's (random) measure, μ_ω^∞, is then defined as limit of $\mu_\omega^{(n)}$ as $n \to \infty$. It is easily verified that $\mu_\omega = \mu_\omega^\infty$.

Kahane and Peyrière (1976) studied the non-degeneration and moments of Z, and gave the Hausdorff dimension of the random measure μ_ω. Guivarc'h (1990) investigated the equation (E), and gave an equivalence of the tail probabilities $P(Z > x)$ as $x \to \infty$ in the non-lattice case. Their works were motivated by questions raised by Mandelbrot (1974) for study of turbulence. Closely related results are given in Kahane (1987), Ben Nasr (1987), Holley and Waymire (1992), Collet and Koukiou (1992), Franchi (1993), Waymire and Williams (1995), Liu (1993, 1996b) and Barral (1997).

To investigate invariant measures of some infinite particle systems, Holley and Liggett (1981) studied the martingale $\{Y_n\}$ and the solutions of (E) in the case where $N = c$ is a constant $\geqslant 2$, and the A_i's $(1 \leqslant i \leqslant c)$ are fixed multiples of one random variable, and Durrett and Liggett (1983) considered the more general case where N is constant but the A_i's $(1 \leqslant i \leqslant c)$ have arbitrary joint distribution.

Example 1.2 (The Galton–Watson process).
(a) $A_1 = \cdots = A_N$ is constant. This is the case where $1 < m = EN < \infty$ and $A_i = 1/m$ if $(1 \leqslant i \leqslant N)$. Therefore $A_u = 1/m$ for all $u \in T(\omega)$, and

$$Y_n = \frac{\operatorname{card}(z_n)}{m^n} \quad (n \geqslant 1)$$

becomes the well-known martingale in the theory of branching processes, $\operatorname{card}(z_n)$ being the population size at nth generation of a Galton–Watson process with offspring distribution given by N. Similar martingales arise in general age-dependent branching processes. Many authors have contributed to the subject, see for example Harris (1948), Kesten and Stigum (1966), Seneta (1968, 1969), Athreya (1971), Athreya and Ney (1972), Doney (1972, 1973) and Bingham and Doney (1974, 1975).

It is also easily seen that the measure μ_ω becomes the "*branching measure*" on $\partial T(\omega)$: we have, for all $u \in T(\omega)$,

$$\mu_\omega(B(u)) = Z(\omega) \lim_{n\to\infty} \frac{\operatorname{card}\{v \in T(\omega)\colon u \leqslant v, |v| = n\}}{\operatorname{card}\{v \in T(\omega)\colon |v| = n\}}.$$

(b) Assume $N \geqslant 1$ a.s. and let $A_{ui} = 1/N_u$ for all $u \in T(\omega)$ and all $1 \leqslant i \leqslant N_u$. The random measure μ_ω is then the "*equally splitting measure*" on $\partial T(\omega)$, in that for all

$u = u_1 \ldots u_n \in T(\omega)$,

$$\mu_\omega(B(u)) = \prod_{i=0}^{n-1} \frac{1}{N_{u|i}}, \quad n \geqslant 1.$$

This measure will be denoted by λ_ω. It corresponds to the Lebesgue measure on $[0,1]$.

The two random measures μ_ω and λ_ω (defined above) have been well studied, see for example Hawkes (1981), Lyons et al. (1995), Liu and Rouault (1996).

Example 1.3 (Self-similar cascades on the Galton–Watson tree). $N \geqslant 1$ a.s. and $A_i = W_i/N$ if $1 \leqslant i \leqslant N, \{W_i\colon i \geqslant 1\}$ being a sequence of i.i.d. non-negative random variables with $EW_1 = 1$, which are also independent of N. Let $\{N_u\colon u \in \boldsymbol{U}\}$ and $\{W_u\colon u \in \boldsymbol{U}\}$ be two independent family of non-negative i.i.d. random variables with $N = N_\emptyset \in \mathbb{N}^\star$ a.s. and $EW_1 = 1$. Let λ_ω be the equally splitting measure on $\partial T(\omega)$ defined as in Example 1.2, and let $\mu_\omega^{(n)}$ be a Borel measure on $\partial T(\omega)$ such that on each $B(u)$ with $u \in z_n(\omega)$,

$$\mu_\omega^{(n)} \text{ has a density } W_{u_1} \ldots W_{u_1 \ldots u_n}$$

with respect to λ_ω. Put $A_{ui} = W_{ui}/N_u$ if $u \in T(\omega)$ and $1 \leqslant i \leqslant N_u$. It is then easily seen that

$$\mu_\omega = \lim_{n \to} \mu_\omega^{(n)}.$$

This is a natural generation of the Mandelbrot's measure to the Galton–Watson tree. Peyrière (1977) constructed such a measure on $[0,1]$ rather than on $\partial T(\omega)$.

Example 1.4 (The branching random walk). The law of $(N, A_1, A_2, \ldots)$ is arbitrary. An initial particle, $\emptyset$, is placed at 0 of the real line $\mathbb{R}$. It gives birth to $N = N_\emptyset(\omega) \geqslant 0$ new particles, u, with displacements $B_u, 1 \leqslant u \leqslant N$. The random variable $(N, B_1, \ldots, B_N)$ can be considered as a point process on $\mathbb{R}$. Each of these new particles, u, behaves independently each other, and gives birth to N_u particles, denoted ui, with displacements B_{ui} from the position of $u, 1 \leqslant i \leqslant N_u$, the random variable $(N_u, B_{u1}, \ldots, B_{uN_u})$ having the same law as $(N, B_1, \ldots, B_N)$. The process continues in a like manner. If $u = u_1 \ldots u_n$ is a particle in nth generation, then its position on $\mathbb{R}$ is given by

$$S_u = B_{u_1} + \cdots + B_{u_1 \ldots u_n}, \quad n \geqslant 1.$$

Assume $1 < EN$ and let $\alpha \in \mathbb{R}$ be such that $m(\alpha) := E\sum_{i=1}^N \mathrm{e}^{-\alpha B_i} < \infty$. As usual, let $T(\omega)$ be the family tree of the process, and let $z_n(\omega)$ be the set of individuals in nth generation. Then the sequence

$$W_n^{(\alpha)} := m(\alpha)^{-n} \sum_{u \in z_n(\omega)} \mathrm{e}^{-\alpha S_u}, \quad n \geqslant 1,$$

is the natural martingale well-studied in the theory of branching random walk, see for example Biggins (1977, 1979), Biggins and Kyprianou (1997), and Chauvin and Rouault (1996). We remark that the martingale $\{W_n^{(\alpha)}\}$ is just our martingale $\{Y_n\}$ with

$$A_u = m(\alpha)^{-1} \mathrm{e}^{-\alpha B_u} \quad \text{for all } u \in T(\omega) \setminus \{\emptyset\}.$$

The martingale (Y_n) and Eq. (E), in its various forms, were also used to study some fractal sets or flows in networks, implicitly or directly by Mauldin and Williams (1986), Falconer (1986, 1987) and Liu (1993, 1996a). Recently, Rösler (1992) and Liu (1996b, 1997a) studied the distributional equation (E) in a general setting.

Let us end this section with some further notations. If E is a set or a statement, we write 1_E or $1\{E\}$ for its indicator function, and card(E) for its cardinality. If X is a random variable, we write $L(X)$ or P_X for its probability law, and we define $||X||_p = E(|X|^p)$ if $p \in (0,1)$, $||X||_p = [E(|X|^p)]^{1/p}$ if $p \in [1,\infty)$, and $||X||_\infty =$ ess sup $|X|$. As usual, we write $f(t) \sim g(t)$ to mean that $\lim_t f(t)/g(t) = 1$, and we put $\log^+ x = \log x$ if $x \geqslant 1$, and $\log^+ x = 0$ otherwise.

2. Main results

We shall always use the convention that $0 \log 0 = 0$. Recall that Z is the a.s. limit of $\{Y_n\}_n$ and that $Y_1 = \sum_{i=1}^N A_i$ by our notations.

The following results have been known.

Theorem 2.0 (Non-degeneration of Z; solution of (E)).

(i) *Assume that* $E|\sum_{i=1}^N A_i \log A_i| < \infty$. *Then* $EZ = 1$ *if*

$$E[Y_1 \log^+ Y_1] < \infty \quad \textit{and} \quad -\infty < E\sum_{i=1}^N A_i \log A_i < 0, \tag{2.1}$$

and $Z = 0$ *a.s. otherwise.*

(ii) *Assuming only that* $1 < E\tilde{N}$ (*which is not necessarily finite*) *and* $EY_1 = 1$. *Then the equation* (E) *has always a non-trivial solution, and there is at most one solution with mean* 1.

Here, we only consider solutions of (E) in the class of probability laws on $[0,\infty)$. It will be useful to remark that the conditions in (2.1) imply the integrability of $\sum_{i=1}^N A_i \log A_i$.

Part (i) is due to Kesten and Stigum (1966) if $A_1 = \cdots = A_N =$ constant, due to Doney (1972) if $A_i \leqslant 1$ $(1 \leqslant i \leqslant N)$, to Kahane and Peyrière (1976) if $N = c \geqslant 2$ is constant and A_i $(1 \leqslant i \leqslant c)$ are i.i.d., and to Durrett and Liggett (1983) if N is constant; it was proved by Biggins (1977) if $\sum_{i=1}^N A_i (\log^+ A_i)^2 < \infty$, by Liu (June 1994, see Liu, 1997a) if $EN < \infty$ and by Lyons (July 1994, see Lyons, 1996) with no additional condition. For Part (ii), the existence was shown in Liu (1998, Theorem 1.1); for a proof of the uniqueness, see Liu (1997a, Corollary 4.2).

The following results will be shown. In Theorems 2.1–2.3 below, we assume (2.1) and $P(Y_1 = 1) < 1$. Thus Z is not a.s. constant.

Theorem 2.1 (Moments). *For each fixed* $p > 1$, $E(Z^p) < \infty$ *if and only if*

$$E[Y_1^p] < \infty \quad \textit{and} \quad E\left[\sum_{i=1}^N A_i^p\right] < 1.$$

Several authors obtained this result in special cases and in very different contexts. For example, it was proved by Kahane and Peyrière (1976) for Mandelbrot's cascades (where N is a constant $\geqslant 2$ and $A_1,\ldots,A_N$ are i.i.d.); by Bingham and Doney (1974, 1975) for the Galton–Watson process and the Crump–Mode–Jagers process (where $A_i \leqslant 1$ a.s. for all i); by Durrett and Liggett (1983) for smoothing processes (where N is a constant $\geqslant 2$), and by Biggins (1979, p. 26; Biggins, 1992, p. 139) for $p \in (1,2)$ in the context of branching random walks. Mauldin and Williams (1986, Theorem 2.1) also gave a proof for the sufficiency of the conditions when $p \in \{2,3,\ldots\}$ and $A_i \leqslant 1$ for all i in the context of random fractals. The case where $p \in \{2,3,\ldots\}$ was also considered by Rösler (1992, Theorems 9 and 10) for a little more general equation where the right side of (E) has an extra term C independent of $\{Z_i\}$. Using the method of Kahane and Peyrière (1976) and that of Bingham and Doney (1975), Liu (1997a) obtained the conclusion in one of the following cases: (i) $p \in (1,2)$ or $p = 2,3,\ldots$, (ii) $||N||_\infty < \infty$, (iii)$||\max_{1\leqslant i\leqslant N} A_i||_\infty < \infty$. To remove these extra conditions, a new method will be applied.

We remark that if $E[\sum_{i=1}^N A_i^p] < 1$ for all $p > 1$, then $||\sup_{1\leqslant i\leqslant N} A_i||_p \leqslant 1$ for all $p > 1$, and so $||\sup_{1\leqslant i\leqslant N} A_i||_\infty \leqslant 1$. Conversely, if a.s. $A_i \leqslant 1$ for all $1\leqslant i\leqslant N$, then the function $\tau(x) = \sum_{i=1}^N A_i^x$ is decreasing in $x \in [1,\infty)$; it is in fact strictly decreasing by (2.1) because $\tau'(x) = E\sum_{i=1}^N A_i^x \log A_i < 0$, $x \geqslant 1$, remarking that $\tau'(x) = 0$ if and only if $P(N = 0,$ or $A_i \in \{0,1\}\ \forall i \in \{1,\ldots,N\}) = 1$, which cannot occur since $\tau'(1) < 0$. Therefore, by Theorem 1 we obtain

Corollary 2.1. *Z has moments of all orders if and only if*

$$P(\forall i \in \{1,\ldots,N\},\ A_i \leqslant 1) = 1 \quad \textit{and} \quad E[Y_1^p] < \infty \textit{ for all } p > 1.$$

The cascade will be called *lattice* if for some $h > 0$ and almost all $\omega \in \Omega$, each $\log A_i$ is an integer multiple of h whenever $1\leqslant i\leqslant N$ and $A_i > 0$; the largest such h will be called the span. Otherwise, it is called *non-lattice*.

Theorem 2.2 (Tail probabilities). *Suppose that for some $\chi > 1$,*

$$E\left[\sum_{i=1}^N A_i^\chi\right] = 1,\ E\left[\sum_{i=1}^N A_i^\chi \log^+ A_i\right] < \infty \quad \textit{and} \quad E\left[\left(\sum_{i=1}^N A_i\right)^\chi\right] < \infty.$$

If the cascade is non-lattice, then the limit $\lim_{x\to\infty} x^\chi P(Z > x)$ exists and is strictly positive and finite; if the cascade is lattice, then

$$0 < \liminf_{x\to+\infty} x^\chi P(Z > x) \leqslant \limsup_{x\to+\infty} x^\chi P(Z > x) < \infty.$$

The theorem is applicable in the case where $P(\forall i \in \{1,\ldots,N\},\ A_i \leqslant 1) < 1$. In the contrary case, the situation differs according to $||N||_\infty < \infty$ or $= \infty$. If $||N||_\infty < \infty$, then $P(Z > x)$ decreases exponentially: the following result was proved by Liu (1996b). Assume $||N||_\infty < \infty$ and that for some $x > 0$, $||\sum_{i=1}^N A_i^x||_\infty \leqslant 1$. Let ρ be the least solution in $(1,\infty)$ of the equation $||\sum_{i=1}^N A_i^\rho||_\infty = 1$. (Such a solution certainly exists under the preceding conditions.) Assume also that for some constants $0 < \delta < 1$, $0 \leqslant a < \infty$, $0 < C < \infty$ and all $0 < x < 1$ sufficiently small, $P(\sum_{i=1}^N A_i^\rho > 1 - x$ and

$A_i \leqslant \delta$ for all $1 \leqslant i \leqslant N) \geqslant Cx^a$. Then for some constants $0 < c_1 \leqslant c_2 < \infty$ and all $x > 0$ sufficiently large,

$$\exp\{-c_2 x^{\rho/(\rho-1)}\} \leqslant P(Z \geqslant x) \leqslant \exp\{-c_1 x^{\rho/(\rho-1)}\}.$$

If $\|N\|_\infty = \infty$, then the tails of Z behave as those of Y_1: see Bingham and Doney (1975).

Theorem 2.2 is due to Guivarc'h if $N = c \geqslant 2$ is constant and A_i $(1 \leqslant i \leqslant c)$ are i.i.d. The proof of Theorems 2.1 and 2.2 develops an idea of Guivarc'h (1990), which considers a random difference equation satisfied by the probability measure $xP_Z(\mathrm{d}x)$. In the canonical case of Mandelbrot, Guivarc'h obtained such an equation by an algebra calculation of measures; this calculation is no longer possible in the general case because of the possible dependence of $\{A_n\}$. To overcome this difficulty, we shall use a natural decomposition of Z using Peryère's measure (Section 4).

The following result gives a necessary and sufficient condition for Z to have an analytic characteristic function.

Theorem 2.3 (Exponential moments). *The following assertions are equivalent*: (i) *a.s.* $A_i \leqslant 1$ *for all* $1 \leqslant i \leqslant N$, *and* $E\mathrm{e}^{tY_1} < \infty$ *for some* $t > 0$; (ii) $E\mathrm{e}^{sZ} < \infty$ *for some* $s > 0$.

The implication (i) $\Rightarrow$ (ii) is a direct consequence of Theorem 6 of Rösler (1992).

To state our next result let us recall the notions of Hausdorff measure and packing measure. (See for example Taylor and Tricot, 1985.) Let (E, d) be a metric space and g be a measure function, that is, a continuous and non-decreasing function defined on $[0, a]$ for some $a > 0$ such that $g(0) = 0$. The *g-Hausdorff* (*outer*) *measure* of $A \subset E$ is defined as

$$g\text{-}H(A) = \lim_{\delta \to 0+} \inf \left\{ \sum_{i=1}^{\infty} g(|U_i|) \colon A \subset \bigcup_{i=1}^{\infty} U_i,\ |U_i| \leqslant \delta \right\},$$

where $|U_i|$ represents the diameter of U_i. If we use covers of just *balls*, we obtain the *spherical g-Hausdorff measure*. On $(\mathbf{I}, d_c)$, the spherical g-Hausdorff measure coincides with the ordinary g-Hausdorff measure, see Liu (1996a, Lemma 0). The *g-packing measure* of A is defined as

$$g\text{-}P(A) = \inf \left\{ \sum_{i=1}^{\infty} g\text{-}\bar{P}(A_i) \colon A \subset \bigcup_{i=1}^{\infty} A_i \right\},$$

where, for each $A \subset E$,

$$g\text{-}\bar{P}(A) = \lim_{\delta \to 0+} \sup \left\{ \sum_{i=1}^{\infty} g(|B_i|) \colon B_i \text{ are disjoint balls with center in } A \text{ and diameter } |B_i| \leqslant \delta \right\}.$$

If $g(t) = t^a$, we write $a\text{-}H(A)$ for $g\text{-}H(A)$, and $a\text{-}P(A)$ for $g\text{-}P(A)$. The Hausdorff dimension of A is defined as

$$\dim(A) = \sup\{a > 0 \colon a\text{-}H(A) = \infty\} = \inf\{a > 0 \colon a\text{-}H(A) = 0\}.$$

The packing dimension Dim(A) is defined in a similar way. To study the Hausdorff measures and packing measures of the support of μ_ω, for simplicity, we assume

$$E[Y_1(\log^+ Y_1)^2] < \infty, \quad E\sum_{i=1}^N A_i(\log A_i)^2 < \infty \quad \text{and} \quad E\sum_{i=1}^N A_i \log A_i < 0, \quad (2.2)$$

and we write

$$\alpha = -E\sum_{i=1}^N A_i \log A_i/\log c \quad \text{and} \quad \sigma^2 = E\sum_{i=1}^N A_i (\log A_i)^2 \left(E\sum_{i=1}^N A_i \log A_i\right)^2. \quad (2.3)$$

Liu and Rouault (1996) showed that α is a.s. the Hausdorff dimension of μ_ω, in that μ_ω is supported by a Borel set K with dimension α, and all Borel set with dimension $< \alpha$ is of μ_ω measure 0. This just says that the measure μ_ω is almost surely supported by a Borel set K with

$$b\text{-}H(K) = \infty \text{ if } 0 < b < \alpha \quad \text{and} \quad b\text{-}H(K) = 0 \text{ if } b > \alpha;$$

and all Borel set $B \subset \partial T(\omega)$ with $b\text{-}H(B) = 0$ for some $b < \alpha$ is of μ_ω measure 0. [That is, a.s. on $\partial T(\omega)$, $\mu_\omega(\cdot) \ll b\text{-}H(\cdot)$ for all $b < \alpha$.] To get more precise results, let us distinguish 2 cases:
(a) for some constant $0 < a < \infty$, a.s. $A_i = 0$ or a for all $1 \leqslant i \leqslant N$;
(b) the contrary case.
We shall see that $\sigma^2 = 0$ in case (a), and $\sigma^2 > 0$ in case (b): cf. Lemma 5.1.

In case (a), the constant a is necessarily $1/E\tilde{N}$, where $\tilde{N} = \sum_{i=1}^N 1\{A_i > 0\}$ is the number of non-zero terms of $A_i, 1 \leqslant i \leqslant N$. This is just the simple Galton–Watson case. By a theorem of Liu (1996a), it is then known that

$$\phi\text{-}H(\partial T) = r^\beta Z,$$

where $\beta = 1 - (\log E\tilde{N}/\log ||\tilde{N}||_\infty)$ ($\beta = 1$ if $||\tilde{N}||_\infty = \infty$), $\phi(t) = t^\alpha(\log^+\log^+(1/t))^\beta$ with $\alpha = \log E\tilde{N}/\log c$, and $r = \sup\{t \geqslant 0\colon E(e^{tZ^{1/\beta}}) < \infty\} \in [0, \infty]$. Of course, this implies that the measure μ is a.s. supported by a Borel set with ϕ-measure strictly positive and finite if $0 < r < \infty$ (which is the case when either $||\tilde{N}||_\infty < \infty$ or $Ee^{t\tilde{N}} < \infty$ for some but not all $t > 0$: see Liu 1996a).

We therefore only consider the case (b).

Theorem 2.4 (Hausdorff measures of support of μ_ω). *Assume* (2.2) *and that there does not exist a constant $a \in (0, \infty)$ such that a.s. $A_i = 0$ or a for all $1 \leqslant i \leqslant N$. Let σ^2 be defined by* (2.3). *Write $\gamma = \sqrt{2\sigma^2/\log c}$ and*

$$\psi_b(t) = t^\alpha e^{b\sqrt{\log^+(1/t)\log^+\log^+\log^+(1/t)}}, \quad b \in (-\infty, \infty).$$

Then $0 < \gamma < \infty$, and, for almost all $\omega \in \Omega$, the measure μ_ω is supported by a Borel set K with

$$\psi_b\text{-}H(K) = \begin{cases} \infty & \text{if } b > \gamma, \\ 0 & \text{if } b < \gamma, \end{cases} \quad \text{and } \psi_b\text{-}P(K) = \begin{cases} \infty & \text{if } b > -\gamma, \\ 0 & \text{if } b < -\gamma; \end{cases}$$

consequently, writing $\xi_b(t) = t^\alpha(\log^+ \frac{1}{t})^b$, $-\infty < b < \infty$, *we have*[1]

$$\alpha\text{-}H(K) = \xi_b\text{-}H(K) = 0 \textit{ for all } b > 0 \textit{ and } \alpha\text{-}P(K) = \xi_b\text{-}P(K) = \infty \textit{ for all } b < 0.$$

Moreover, almost surely, all Borel sets B with $\psi_b\text{-}H(B) = 0$ *for some* $b > \gamma$ *is of* μ_ω*-measure* 0. (*That is, a.s.* $\mu_\omega \ll \psi_b\text{-}H$ *for all* $b > \gamma$.)

For applications of the theorem to examples of Section 1, the following observations are useful. In Example 1.1,

$$\sigma^2 = EW_1(\log W_1)^2 - (E\ W_1 \log W_1)^2$$

and $\sigma^2 = 0$ if and only if for some constant $a > 0$, $W_1 = 0$ or a a.s. In Example 1.2(b),

$$\sigma^2 = E(\log N)^2 - (E \log N)^2,$$

and $\sigma^2 = 0$ if and only if for some constant $c \in \mathbb{N}^\star$, $N = c$ a.s. In Example 1.3,

$$\sigma^2 = EW_1(\log W_1)^2 - (EW_1 \log W_1)^2 + E(\log N)^2 - (E \log N)^2,$$

and $\sigma^2 = 0$ if and only if for some constants $a > 0$ and $c \in \mathbb{N}^\star$, $W_1 = 0$ or a and $N = c$ a.s. An another example, equally interesting, is the case where given $N = n \geqslant 2$, $\{A_i\}_{1 \leqslant i \leqslant n}$ are i.i.d., and their conditional law is the law of A_1. Then

$$\sigma^2 = (EN)E[A_1(\log A_1)^2] - [(EN)EA_1 \log A_1)]^2,$$

and $\sigma^2 = 0$ if and only if a.s., $A_1 = 0$ or $1/[(EN)P(A_1 > 0)]$.

We have already seen that in case (a) and under some conditions, there is a gauge ϕ for which μ is a.s. supported by a set with ϕ-measure strictly positive and finite. We believe that in case (b), such a gauge would not exist. However, if we consider $d(u, v) = X_{u\wedge v}$ instead of the distance d_c, we can prove a result which is similar to that in the case (a): cf. Liu (1993).

In the case where $N = c$ is a constant $\geqslant 2$ and $A_1, \ldots, A_c$ are independent and identically distributed, the dimension result is due to Kahane and Peyrière (1976); it was also shown, by Kahane (1987), that μ_ω is a.s. supported by a Borel set K with $g\text{-}H(K) = 0$ if g is a measure function satisfying $\int_0^1 g(t)t^{-\alpha-1}\,\mathrm{d}t < \infty$. (This integral condition holds for example for $g(t) = t^\alpha(\log(1/t))^{-(1+\varepsilon)}$ with $\varepsilon > 0$, which is a function *smaller then* t^α.) We remark that Kahane's result cannot imply $\alpha\text{-}H(K) = 0$.

It was shown, by Liu (1998), that the functional function (E′) has always non-trivial solutions, even if Z is degenerate. What these solutions stand for? The next theorem shows that, in the case where Z is degenerate, there are other natural weighted martingales converging to non-degenerate random variables and leading to non-trivial solutions of (E′).

Theorem 2.5 (Normalized martingales in the case where $Z = 0$ a.s.).
(a) *If* $E\sum_{i=1}^N A_i \log A_i = 0$, *then*

$$Y_n^* = \sum_{|u|=n, u\in T(\omega)} X_u \log\frac{1}{X_u}, \quad n \geqslant 1,$$

[1] It suffices to remark that $\lim_{t\to 0} \frac{\psi_b(t)}{\xi_{b_0}(t)} = \begin{cases} 0 & \text{if } b_0 > 0 \text{ and } b > 0, \\ \infty & \text{if } b_0 < 0 \text{ and } b < 0. \end{cases}$

is a martingale (associated with the filtration $\{\mathscr{F}_n\}$), where the sum is taken over all nodes u of $T(\omega)$ with length n such that $X_u > 0$. If additionally,

$$\text{for some } \delta > 0,\ E\tilde{N}^{1+\delta} < \infty \text{ and } E\left(\sum_{i=1}^{N} A_i\right)^{1+\delta} < \infty,$$

then for some non-degenerate random variable $Z^* \geqslant 0$,

$$\lim_{n\to\infty} Y_n^* = Z^* \quad P\text{-a.s.}$$

Moreover, the Laplace transform of Z^*, $\phi^*(t) = E\mathrm{e}^{-tZ^*}$, $t \geqslant 0$, is a (non-trivial) solution of (E′) satisfying

$$\lim_{t\to 0+} \frac{1-\phi^*(t)}{t|\log t|} = 1.$$

(b) *If $E\sum_{i=1}^{N} A_i \log A_i > 0$, then*

$$Y_n^{(\alpha)} = \sum_{|u|=n, u\in T(\omega)} X_u^{\alpha}, \quad n \geqslant 1,$$

is a martingale (associated with the filtration $\{\mathscr{F}_n\}$), where α is the unique number in $(0,1)$ satisfying $E\sum_{i=1}^{N} A_i^{\alpha} = 1$. If additionally,

$$E\left(\sum_{i=1}^{N} A_i^{\alpha}\right) \log^{+}\left(\sum_{i=1}^{N} A_i^{\alpha}\right) < \infty,$$

then for some non-degenerate random variable $Z_\alpha \geqslant 0$,

$$\lim_{n\to\infty} Y_n^{(\alpha)} = Z_\alpha \quad P\text{-a.s.}$$

Moreover, the function $\phi_\alpha(t) = E\mathrm{e}^{-t^{\alpha}Z_\alpha}$, $t \geqslant 0$, is a Laplace transform of some probability distribution on $[0,\infty)$, and is a (non-trivial) solution of (E′) satisfying

$$\lim_{t\to 0+} \frac{1-\phi_\alpha(t)}{t^{\alpha}} = 1.$$

We remark that the martingale $\{Y_n^*\}$ satisfies $EY_n^* = 0$ $(n \geqslant 1)$; it is in general not non-negative. However, since $\lim_{n\to\infty} \max_{|u|=n, u\in T(\omega)} X_u = 0$ a.s., we see that for almost all ω, there is $n_0(\omega) \in \mathbb{N}$ such that $Y_n^*(\omega) \geqslant 0$ for all $n \geqslant n_0(\omega)$. This is different to saying that for some $n_0 > 0$ sufficiently large and all $n > n_0$, $Y_n^*(\omega) \geqslant 0$ a.s. It would be interesting to prove the a.s. convergence of $\{Y_n^*\}$ without assuming the moment conditions of order $1+\delta$.

3. The random difference equation

In this section, $(\Omega, \mathscr{F}, P)$ denote an arbitrary probability space, (A,B) and (A_n, B_n) $(n \geqslant 1)$ are i.i.d. random variable defined on $(\Omega, \mathscr{F}, P)$, with values in $\mathbb{R}^2$.

Consider the random difference equation

$$X \stackrel{\mathrm{L}}{=} AX + B, \tag{3.1}$$

where X is a real random variable independent of (A,B), and $\overset{\mathrm{L}}{=}$ denotes equality in law; the law of X is unknown. In terms of characteristic functions, the equation reads

$$\phi(t) = E[\mathrm{e}^{\mathrm{i}Bt}\phi(At)], \quad t \in R. \tag{3.1'}$$

A probability law, μ, is said to be a solution of (3.1) if there is a random variable X having μ as its law and satisfying (3.1); when we say that a random variable X is the unique solution of (3.1), we mean that the corresponding law is the unique solution.

Lemma 3.1 (Grintsevichyus, 1974, Theorem 1 and Proposition 1)**.** *If*

$$P(A \neq 0) = 1, \quad -\infty < E\log|A| < 0 \quad \textit{and} \quad E\log^+|B| < \infty, \tag{3.2}$$

then (3.1) *has a unique solution, and this solution is given by*

$$X = B_1 + A_1B_2 + A_1A_2B_3 + \cdots + A_1\cdots A_{n-1}B_n + \cdots \tag{3.3}$$

the series being convergent a.s.

Lemma 3.2. *Assume* (3.2) *and let X be the unique solution of* (3.1). *Let p be any fixed number in* $(0,\infty)$. *If*

$$E(|A|^p) < 1 \quad \textit{and} \quad E(|B|^p) < \infty, \tag{3.4}$$

then $E(|X|^p) < \infty$; *the converse holds if additionally $A \geqslant 0$ and $B \geqslant 0$ a.s. with* $P(B > 0) > 0$.

Proof. By Lemma 3.1, we can suppose that X is given by (3.3). If $||A||_p < 1$ and $||B||_p < \infty$, then by (3.3) and the triangular inequality for $||.||_p$, we obtain

$$||X||_p \leqslant ||B||_p + ||A||_p||B||_p + \cdots + ||A||_p^{n-1}||B||_p + \cdots = \frac{||B||_p}{1 - ||A||_p} < \infty.$$

Conversely if $A \geqslant 0$ and $B \geqslant 0$ a.s. and if X is a solution of (3.1) with $E(|X|^p) < \infty$, then $X \geqslant 0$ by (3.3), and $E(B^p) \leqslant E[(AX+B)^p] < \infty$ by (3.1); since $P(B > 0) > 0$, we have also $P(X > 0) > 0$, $0 < E(X^p) < \infty$ and

$$E[(AX)^p] < E[(AX+B)^p] = E(X^p),$$

which implies $E[(A)^p] < 1$ by the independence of A and X. $\square$

Lemma 3.3 (Kesten, 1973; Grintsevichyus, 1975)**.** *Assume that* $P(A > 0) = P(B \geqslant 0) = 1$, $P(B > 0) > 0$, *and that for some* $\lambda \in (0,\infty)$,

$$E(A^\lambda) = 1, E(A^\lambda \log^+ A) < \infty \textit{ and } E(B^\lambda) < \infty.$$

Let X be the unique solution of (3.1) *and suppose that X is not a.s. a constant.* [*That is, there does not exist a constant c such that* $P(c = Ac + B) = 1$.] *Then*

(a) *if* $\log A$ *is not of lattice type* (*that is, there does not exist $h > 0$ such that a.s.* $\log A$ *is an integer multiple of h*), *then for some constant* $d \in (0,\infty)$,

$$\lim_{x\to\infty} x^\lambda P(X > x) = d; \tag{3.5}$$

(b) *if* $\log A$ *is of lattice type with span* $h>0$, *then for each* $x\in\mathbb{R}$, *there is* $d(x)\in(0,\infty)$ *such that*

$$\lim_{n\to\infty} \mathrm{e}^{(x+nh)\lambda}P(X>\mathrm{e}^{x+nh})=d(x). \tag{3.6}$$

In the non-lattice case, the result was proved by Kesten (1973); it was also proved independently by Grintsevichyus (1975) under an additional condition. A simple proof was given by Guivarc'h (1990). For the lattice case, see Grintsevichyus (1975).

4. The random difference equation satisfied by $xP_Z(\mathrm{d}x)$; proofs of Theorems 2.1–2.3

For $u\in\boldsymbol{U}$, we use the notations A_u, X_u and Z_u introduced in Section 1. For $(\omega,\boldsymbol{i})\in\Omega\times\boldsymbol{I}$, put

$$\tilde{Z}(\omega,\boldsymbol{i})=Z_\emptyset(\omega)=Z(\omega),$$
$$\tilde{A}_1(\omega,\boldsymbol{i})=A_{\boldsymbol{i}|1}(\omega),$$
$$\tilde{Z}_1(\omega,\boldsymbol{i})=Z_{\boldsymbol{i}|1}(\omega).$$

They are measurable functions on $\Omega\times\boldsymbol{I}$ associated with the product σ-field of $\mathscr{F}$ and $\mathscr{B}$, $\mathscr{B}$ being the Borel σ-field on $\boldsymbol{I}$. Let Q be the Peyrière's measure on $\Omega\times\boldsymbol{I}$, defined by

$$Q(A)=E\int_{\boldsymbol{I}} 1_A(\omega,\boldsymbol{i})\mu_\omega(\mathrm{d}\boldsymbol{i}),\quad A\in\mathscr{F}\times\mathscr{B}.$$

As $EZ=1$, Q is a probability measure. We write $E_Q[f]$ for the integration of f with respect to Q. The following results give the distributions of the random variables $\tilde{Z},(\tilde{A}_1,\tilde{B}_1)$, and shows that the probability law $xP_Z(\mathrm{d}x)$ satisfies a random difference equation. Let us recall the convention that the empty sum is taken to be 0.

Lemma 4.1. *For all* $(\omega,\boldsymbol{i})\in\Omega\times\boldsymbol{I}$,

$$\tilde{Z}(\omega,\boldsymbol{i})=\tilde{A}_1(\omega,\boldsymbol{i})\tilde{Z}_1(\omega,\boldsymbol{i})+\tilde{B}_1(\omega,\boldsymbol{i}),$$

where

$$\tilde{B}_1(\omega,\boldsymbol{i})=\sum_{i=1}^{N(\omega)}A_i(\omega)Z_i(\omega)1\{(\boldsymbol{i}|1)\neq i\},$$

$\tilde{Z}_1$ *is* Q*-independent of* $(\tilde{A}_1,\tilde{B}_1)$, *and has the same distribution as* $\tilde{Z}$. *Moreover, for all non-negative Borel functions* f,g *and* h (*defined on* $\mathbb{R}$ *or* $\mathbb{R}^2$),

$$E_Q[f(\tilde{A}_1)]=E\left[\sum_{i=1}^{N}f(A_i)A_i\right],$$

$$E_Q[f(\tilde{B}_1)]=E\left[\sum_{1\leqslant k\leqslant N}A_k f\left(\sum_{1\leqslant i\leqslant N,i\neq k}A_iZ_i\right)\right],$$

$$E_Q[h(\tilde{A}_1,\tilde{B}_1)] = E\left[\sum_{1\leqslant k\leqslant N} A_k h\left(A_k, \sum_{1\leqslant i\leqslant N, i\neq k} A_i Z_i\right)\right],$$

$$E_Q[g(\tilde{Z}_1)] = E[g(Z)Z] = E_Q[g(\tilde{Z})].$$

In particular, $P_{\tilde{Z}}(\mathrm{d}x) = xP_Z(\mathrm{d}x)$.

Proof. We have

$$\begin{aligned} A_{\boldsymbol{i}|1}Z_{\boldsymbol{i}|1} &= A_iZ_i \quad \text{if } (\boldsymbol{i}|1)=i \\ &= \sum_{i=1}^{\infty} A_iZ_i 1\{(\boldsymbol{i}|1)=i\} = \sum_{i=1}^{N} A_iZ_i 1\{(\boldsymbol{i}|1)=i\}, \end{aligned}$$

$$\begin{aligned} Z = \sum_{i=1}^{N} A_iZ_i &= \sum_{i=1}^{N} A_iZ_i[1\{(\boldsymbol{i}|1)=i\} + 1\{(\boldsymbol{i}|1)\neq i\}] \\ &= A_{\boldsymbol{i}|1}Z_{\boldsymbol{i}|1} + \sum_{i=1}^{N} A_iZ_i 1\{(\boldsymbol{i}|1)\neq i\}. \end{aligned}$$

Therefore

$$\tilde{Z}(\omega,\boldsymbol{i}) = \tilde{A}_1(\omega,\boldsymbol{i})\tilde{Z}_1(\omega,\boldsymbol{i}) + \tilde{B}_1(\omega,\boldsymbol{i}).$$

We claim that $\tilde{Z}_1$ is independent of $(\tilde{A}_1,\tilde{B}_1)$, and has the same distribution as $\tilde{Z}$; by the way, we shall give the distribution of $(\tilde{A}_1,\tilde{B}_1)$. In fact, for all non-negative Borel functions g and h, we have

$$\begin{aligned} E_Q[h(\tilde{A}_1,\tilde{B}_1)g(\tilde{Z}_1)] &= E\left[\sum_{1\leqslant k\leqslant N} h\left(A_k, \sum_{1\leqslant i\leqslant N, i\neq k} A_iZ_i\right) g(Z_k)A_kZ_k\right] \\ &= E\left[\sum_{1\leqslant k\leqslant N} A_k h\left(A_k, \sum_{1\leqslant i\leqslant N, i\neq k} A_iZ_i\right)\right] E[g(Z)Z]. \end{aligned}$$

Taking $g=1$ or $h=1$ gives the expressions of $E_Q[h(\tilde{A}_1,\tilde{B}_1)]$ and $E_Q[g(\tilde{Z}_1)]$. Consequently

$$E_Q[h(\tilde{A}_1,\tilde{B}_1)g(\tilde{Z}_1)] = E_Q[h(\tilde{A}_1,\tilde{B}_1)]\ E_Q[g(\tilde{Z}_1)].$$

This gives the independence of $(\tilde{A}_1,\tilde{B}_1)$ and $\tilde{Z}_1$. The expressions of $E_Q[f(\tilde{A}_1)]$ and $E_Q[f(\tilde{B}_1)]$ come from $E_Q[h(\tilde{A}_1,\tilde{B}_1)]$ by taking $h(x,y)=f(x)$ or $f(y)$. So the proof is finished. $\square$

Lemma 4.2. *For all* $p>1$,

$$E_Q[(\tilde{B}_1)^{p-1}] \leqslant \begin{cases} E\left[\left(\sum_{k=1}^{N} A_k\right)^p\right] & \text{if } p\leqslant 2, \\ E[Z^{p-1}]E\left[\left(\sum_{k=1}^{N} A_k\right)^p\right] & \text{if } p>2. \end{cases}$$

Proof. By Lemma 4.1,

$$E_Q[(\tilde{B}_1)^{p-1}] = E\left[\sum_{1\leqslant k\leqslant N} A_k \left(\sum_{1\leqslant i\leqslant N, i\neq k} A_i Z_i\right)^{p-1}\right].$$

Denote by I the expectation above. If $p-1\leqslant 1$, then by the concavity of the function $x\mapsto x^{p-1}$ and Jensen's inequality, for fixed $k, 1\leqslant k\leqslant N$,

$$E\left[\left(\sum_{1\leqslant i\leqslant N, i\neq k} A_i Z_i\right)^{p-1}\middle|\mathscr{F}_1\right] \leqslant \left(E\left[\sum_{1\leqslant i\leqslant N, i\neq k} A_i Z_i \middle| \mathscr{F}_1\right]\right)^{p-1}$$

$$= \left(\sum_{1\leqslant i\leqslant N, i\neq k} A_i\right)^{p-1} \leqslant \left(\sum_{i=1}^{N} A_i\right)^{p-1},$$

so that $I\leqslant E[(\sum_{k=1}^N A_k)^p]$; If $p-1\geqslant 1$, using the inequality

$$\left(\sum a_i z_i\right)^{p-1} \leqslant \sum a_i z_i^{p-1}, \quad a_i\geqslant 0, \ \sum a_i = 1, \ z_i\geqslant 0$$

(the convexity of the function $z\mapsto z^{p-1}$) for $a_i = A_i/\sum_{j\neq k} A_j$ and $z_i = Z_i$, $i\neq k$, we obtain

$$\left(\sum_{i\neq k} A_i Z_i\right)^{p-1} \leqslant \left(\sum_{i\neq k} A_i\right)^{p-2} \left(\sum_{i\neq k} A_i Z_i^{p-1}\right)$$

(if $\sum_{j\neq k} A_j = 0$, the inequality is evident). Consequently

$$I \leqslant E(Z^{p-1}) E\left[\sum_{k=1}^{N} A_k \left(\sum_{i\neq k} A_i\right)^{p-2} \left(\sum_{i\neq k} A_i\right)\right]$$

$$\leqslant E(Z^{p-1}) E\left[\left(\sum_{i=1}^{N} A_i\right)^{p}\right]. \qquad \square$$

The first part of the following result may be considered as a consequence of Theorem 8.1.4 of Bingham et al. (1987, p. 332). For convenience of readers, we present a direct proof, following some ideas of Feller (1971, Chapter III.9).

Lemma 4.3. *Let μ be a probability distribution on $\mathbb{R}$ and, for all $x\geqslant 0$ and all $a\in\mathbb{R}$, put $M_a(x) = \int_x^\infty u^a \mu(du)$ $(\leqslant\infty)$. Then we have the following relations between the tail probability $M_0(x) = \mu(x,\infty)$ and the truncated moments $M_a(x)$, $a\neq 0$:*

(i) *let $\ell(.)$ be a function slowly varying at ∞. Then for each $\rho>0$ and each $a\in\mathbb{R}$ with $\rho - a > 0$, as $x\to\infty$, $M_0(x)\sim x^{-\rho}\ell(x)/\rho$ if and only if $M_a(x)\sim x^{-(\rho-a)}\ell(x)/(\rho-a)$;*

(ii) *if for some* $a>0$, $b>1$, $c>0$, $d>0$ *and* $\rho>0$, $\lim_{n\to\infty}(cb^n)^\rho M_a(cb^n)=d$, *then*

$$\left[1-\frac{1-b^{-a}}{1-b^{-\rho-a}}\right]d\leqslant\liminf_{n\to\infty}\frac{M_0(cb^n)}{(cb^n)^{-(a+\rho)}}$$

$$\leqslant\limsup_{n\to\infty}\frac{M_0(cb^n)}{(cb^n)^{-(a+\rho)}}\leqslant\left[1-\frac{(1-b^{-a})b^{-\rho}}{1-b^{-\rho-a}}\right]d.$$

Proof. (i) We begin with the following identities: for all $x>0$ and all $a\in\mathbb{R}$,

(a) $M_a(x)-x^aM_0(x)=a\displaystyle\int_x^\infty y^{a-1}M_0(y)\,\mathrm{d}y$,

(b) $M_0(x)-x^{-a}M_a(x)=-a\displaystyle\int_x^\infty y^{-a-1}M_a(y)\,\mathrm{d}y$.

Both identities follow from Fubini's theorem, remarking that the right-hand sides can be expressed as double integrals using the definition of M_a.

If $M_0(x)$ varies regularly at ∞ with exponent $-\rho$, we apply Theorem 1 of Feller (1971, p. 281) to the right-hand side of (a); if $M_a(x)$ varies regularly with exponent $-(\rho-a)$, we apply the same theorem to the right-hand side of (b). In both cases, we obtain

$$\frac{M_a(x)}{x^aM_0(x)}\to\frac{\rho}{\rho-a}\quad(x\to\infty),$$

which also holds even if $\rho=0$ or $\rho-a=0$, provided that $\rho/(\rho-a)$ is interpreted to be 0 or ∞ according to $\rho=0$ or $\rho-a=0$, respectively. This implies the desired assertion.

(ii) Denote by $-I(x)$ the right-hand side of (b); then $I(cb^n)=\sum_{k=n}^\infty I_k$, where $I_k=a\int_{cb^k}^{cb^{k+1}}y^{-a-1}M_a(y)\,\mathrm{d}y$. By the monotonicity of $M_a(.)$, $M_a(cb^{k+1})(cb^k)^{-a}(1-b^{-a})\leqslant I_k\leqslant M_a(cb^k)(cb^k)^{-a}(1-b^{-a})$; by the condition, for each $\varepsilon>0$ and all k large enough, $M_a(cb^k)$ lies between $(d\pm\varepsilon)(cb^k)^{-\rho}$. These lead to a lower bound and an upper bound of $I(cb^n)$ for n large enough, so that the conclusion follows from the identity (b) with $x=cb^n$. □

Proof of Theorem 2.1. By Lemma 4.1,

$$E_Q[\tilde{Z}_1^{p-1}]=E[Z^p]\ \text{and}\ E_Q[\tilde{A}_1^{p-1}]=E\left[\sum_{i=1}^N A_i^p\right].$$

So by Lemmas 3.2 and 4.2, we see that for all $p>1$, if

$$E[Z^{p-1}]<\infty,\quad E\left[\left(\sum_{k=1}^N A_k\right)^p\right]<\infty\ \text{and}\ E\left[\sum_{k=1}^N A_k^p\right]<1,$$

then $E[Z^p]<\infty$. Noting that $EZ<\infty$, an easy induction argument on n shows that, for all $n=2,3,\dots$, if

$$p\in(n-1,n],\quad E\left[\left(\sum_{k=1}^N A_k\right)^p\right]<\infty\ \text{and}\ E\left[\sum_{k=1}^N A_k^p\right]<1,$$

then $E[Z^p] < \infty$. This gives the sufficiency of the conditions. The necessity is given in Liu (1997a), and can be obtained as follows. Assume $EZ^p < \infty$. That is, $E_Q(\tilde{Z})^{p-1} < \infty$. So by Lemma 3.2, $E_Q(\tilde{A}_1)^{p-1} < 1$. This just says $E[\sum_{k=1}^N A_k^p] < 1$. On the other hand, by Jensen's inequality, we have $[E(Z|\mathscr{F}_1)]^p \leqslant [E(Z^p|\mathscr{F}_1)]$; by Eq. (E), $E(Z|\mathscr{F}_1) = Y_1$. So

$$EY_1^p = E[E(Z|\mathscr{F}_1)]^p \leqslant E[E[Z^p|\mathscr{F}_1]] = EZ^p < \infty.$$

The proof of Theorem 2.1 is the finished. □

Proof of Theorem 2.2. By Lemma 4.1, we have

$$E_Q[\tilde{A}_1^{\chi-1}] = E\left[\sum_{i=1}^N A_i^\chi\right] = 1 \text{ and } E_Q[\tilde{A}_1^{\chi-1}\log^+\tilde{A}_1] = E\left[\sum_{i=1}^N A_i^\chi \log^+ A_i\right] < \infty;$$

by Theorem 2.1, $EZ^{\chi-1} < \infty$, so that by Lemma 4.2, $E_Q[\tilde{B}_1^{\chi-1}] < \infty$. Write $M_1(t) = Q(\tilde{Z} > t)$; then $M_1(t) = \int_t^\infty x P_Z(\mathrm{d}x)$. If the cascade is non-lattice, then by Lemma 3.3, the limit

$$d = \lim_{t\to+\infty} t^{\chi-1} M_1(t)$$

exists and is strictly positive and finite. By Lemma 4.3(i), this implies

$$\lim_{t\to+\infty} t^\chi P(Z > t) = d(\chi-1)/\chi.$$

If the cascade is lattice with span $h > 0$, then by Lemma 3.3, for each $x \in \mathbb{R}$, the limit

$$d(x) = \lim_{n\to\infty} \mathrm{e}^{(x+nh)(\chi-1)} M_1(\mathrm{e}^{x+nh})$$

exists with $0 < d(x) < \infty$ and $d(x+h) = d(x)$ for all $x \in \mathbb{R}$. Using this for $x = 0$ together with Lemma 4.3(ii), we see that

$$0 < \liminf_{t\to+\infty} \mathrm{e}^{nh\chi} P(Z > \mathrm{e}^{nh}) \leqslant \limsup_{t\to+\infty} \mathrm{e}^{nh\chi} P(Z > \mathrm{e}^{nh}) < \infty,$$

and therefore $0 < \liminf_{t\to+\infty} t^\chi P(Z > t) \leqslant \limsup_{t\to+\infty} t^\chi P(Z > t) < \infty$ by the monotonicity of $P(Z > t)$. □

Proof of Theorem 2.3. By Theorem 6 of Rösler (1992), we need only to prove that (ii) ⇒ (i). Assume (ii). By Jensen's inequality, we have, for all $t \in \mathbb{R}$,

$$\begin{aligned} E\mathrm{e}^{tZ} &= E\mathrm{e}^{t(A_1Z_1+\cdots+A_NZ_N)} \\ &= E[E[\mathrm{e}^{t(A_1Z_1+\cdots+A_NZ_N)}|\mathscr{F}_1]] \\ &\geqslant E[\mathrm{e}^{E[t(A_1Z_1+\cdots+A_NZ_N)|\mathscr{F}_1]}] \\ &= E\mathrm{e}^{t(A_1+\cdots+A_N)} = E\mathrm{e}^{tY_1}. \end{aligned}$$

Therefore for all $t > 0$, $E\mathrm{e}^{tZ} < \infty$ implies $E\mathrm{e}^{tY_1} < \infty$. On the other hand, (ii) implies in particular that Z has finite moments of all orders, so that by Corollary 2.1 a.s. $A_i \leqslant 1$ if $1 \leqslant i \leqslant N$. □

5. Hausdorff and packing measures of the support of μ_ω : Proof of Theorem 2.4

We first give a necessary and sufficient condition for $\sigma^2 = 0$.

Lemma 5.1. *Assume* (2.2). *Then* $\sigma^2 = 0$ *if for some constant* $0 < a < \infty$, *a.s.* $A_i = a$ *whenever* $1 \leqslant i \leqslant N$ *and* $A_i > 0$, *and* $\sigma^2 > 0$ *otherwise.*

Proof. We first remark that σ^2 is the variance of $\log \tilde{A}_1$:

$$\sigma^2 = E_Q(\log \tilde{A}_1)^2 - (E_Q \log \tilde{A}_1)^2.$$

Therefore, $\sigma^2 = 0$ if and only if, for some constant $0 < a < \infty$, $Q(\tilde{A}_1 = a) = 1$. That is,

$$E \sum_{i=1}^{N} A_i 1\{A_i = a\} = 1.$$

Because $E \sum_{i=1}^{N} A_i = 1$, the last statement is equivalent to

$$E \sum_{i=1}^{N} A_i (1 - 1\{A_i = a\}) = 0,$$

which just says that a.s., $A_i = 0$ or a for all $1 \leqslant i \leqslant N$. $\square$

We next give a density theorem about the measure μ_ω.

Lemma 5.2. *Assume* (2.2) *and* $\sigma^2 > 0$. *Then for P-almost all* $\omega \in \Omega$ *and* μ_ω*-almost all* $\boldsymbol{i} \in \boldsymbol{I}$,

$$\limsup_{n\to\infty} \frac{\mu_\omega(B(\boldsymbol{i}|n))}{\psi_b(|B(\boldsymbol{i}|n)|)} = \begin{cases} 0 & \text{if } b > \gamma, \\ \infty & \text{if } b < \gamma. \end{cases}$$

and

$$\liminf_{n\to\infty} \frac{\mu_\omega(B(\boldsymbol{i}|n))}{\psi_b(|B(\boldsymbol{i}|n)|)} = \begin{cases} 0 & \text{if } b > -\gamma, \\ \infty & \text{if } b < -\gamma. \end{cases}$$

Proof. By the definition of μ_ω, we have

$$\mu_\omega(B(\boldsymbol{i}|n)) = X_{\boldsymbol{i}|n} Z_{\boldsymbol{i}|n} = \mathrm{e}^{S_n} \tilde{Z}_n, \quad n \geqslant 0,$$

where $\tilde{Z}_n(\omega, \boldsymbol{i}) = Z_{\boldsymbol{i}|n}(\omega)$,

$$S_0 = 1 \quad \text{and} \quad S_n = \log \tilde{A}_1 + \cdots + \log \tilde{A}_n \quad (n \geqslant 1)$$

with $\tilde{A}_k(\omega, \boldsymbol{i}) = A_{\boldsymbol{i}|k}(\omega)$ $(k \geqslant 1)$. It is known that the sequence $\{\tilde{A}_k\}$ $(k \geqslant 1)$ is Q-independent and identically distributed, with

$$E_Q \log \tilde{A}_1 = E \sum_{i=1}^{N} A_i \log A_i < 0.$$

(see Liu and Rouault, 1996, Lemma 10.) By the iterated logarithm law,

$$\limsup_{n\to\infty} \frac{S_n - E_Q S_n}{\sqrt{2\sigma^2 n \log\log n}} = 1 \quad Q\text{-a.s.}$$

It follows that for Q-almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$ and all $0 < \varepsilon < 1$,

$$e^{nE\sum_{i=1}^{N} A_i \log A_i + (1-\varepsilon)\sqrt{2\sigma^2 n \log\log n}} \leqslant e^{S_n} \leqslant e^{nE\sum_{i=1}^{N} A_i \log A_i + (1+\varepsilon)\sqrt{2\sigma^2 n \log\log n}},$$

where the left inequality holds for infinitely many $n \in \mathbb{N}$, while the right inequality holds for all $n \in \mathbb{N}$ sufficiently large. Moreover, it is known that if $EZ \log^+ Z < \infty$, then

$$\lim_{n\to\infty} \frac{\log \tilde{Z}_n}{n} = 0 \quad Q\text{-a.s.}$$

[see Liu and Rouault (1996, Lemma 12) where the condition $P(Z > 0) = 1$ is not needed.] Since $EY_1(\log^+ Y_1)^2 < \infty$ implies $EZ \log^+ Z < \infty$ (cf. Biggins 1979, Section 4), it follows that for Q-almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$ and all $n \in \mathbb{N}$ sufficiently large,

$$e^{-n\varepsilon} \leqslant \tilde{Z}_n \leqslant e^{n\varepsilon}.$$

Consequently, for Q almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$ and all $0 < \varepsilon < 1/2$,

$$e^{nE\sum_{i=1}^{N} A_i \log A_i + (1-2\varepsilon)\sqrt{2\sigma^2 n \log\log n}} \leqslant e^{S_n}\tilde{Z}_n \leqslant e^{nE\sum_{i=1}^{N} A_i \log A_i + (1+2\varepsilon)\sqrt{2\sigma^2 n \log\log n}},$$

where the left inequality holds for infinitely many $n \in \mathbb{N}$, while the right inequality holds for all $n \in \mathbb{N}$ sufficiently large. Therefore, for Q almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$ and all $0 < \varepsilon < 1/2$,

$$\limsup_{n\to\infty} \frac{e^{S_n}\tilde{Z}_n}{e^{nE\sum_{i=1}^{N} A_i \log A_i + (1+2\varepsilon)\sqrt{2\sigma^2 n \log\log n}}} \leqslant 1$$

and

$$\limsup_{n\to\infty} \frac{e^{S_n}\tilde{Z}_n}{e^{nE\sum_{i=1}^{N} A_i \log A_i + (1-2\varepsilon)\sqrt{2\sigma^2 n \log\log n}}} \geqslant 1.$$

Hence, for Q almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$ and all $0 < \varepsilon < 1/3$,

$$\limsup_{n\to\infty} \frac{e^{S_n}\tilde{Z}_n}{e^{nE\sum_{i=1}^{N} A_i \log A_i + (1+3\varepsilon)\sqrt{2\sigma^2 n \log\log n}}} = 0$$

and

$$\limsup_{n\to\infty} \frac{e^{S_n}\tilde{Z}_n}{e^{nE\sum_{i=1}^{N} A_i \log A_i + (1-3\varepsilon)\sqrt{2\sigma^2 n \log\log n}}} = \infty.$$

Since $|B(\boldsymbol{i}|n)| = c^{-n}$, $|B(\boldsymbol{i}|n)|^\alpha = c^{-n\alpha} = e^{nE\sum_{i=1}^{N} A_i \log A_i}$ and for n large enough, $\psi_b(|B(\boldsymbol{i}|n)|) = c^{-n\alpha} e^{b\sqrt{n(\log c)\log\log(n\log c)}}$, it follows that for Q almost all $(\omega, \boldsymbol{i}) \in (\Omega, \boldsymbol{I})$,

$$\limsup_{n\to\infty} \frac{\mu_\omega(B(\boldsymbol{i}|n))}{\psi_b(|B(\boldsymbol{i}|n)|)} = \begin{cases} 0 & \text{if } b > \gamma, \\ \infty & \text{if } b < \gamma. \end{cases}$$

This gives the desired conclusion for limsup. The proof for liminf is similar, using

$$\liminf_{n\to\infty} \frac{S_n - E_Q S_n}{\sqrt{2\sigma^2 n \log\log n}} = -1 \quad Q\text{-a.s.} \qquad \square$$

The *Proof of Theorem 2.4* is then easy: the result is an immediate consequence of Lemmas 5.1 and 5.2, using classical density theorems on Hausdorff measures and packing measures, see for example Dai and Taylor (1992, Section 5), Taylor and Tricot (1985) or Taylor (1986, Section 4). □

6. Other weighted martingales: Proof of Theorem 2.5

We first verify that $\{Y_n^*\}$ is a martingale. By the branching property, we have for all $n \geqslant 0$,

$$E[Y_{n+1}^* | \mathscr{F}_n] = E\left[\sum_{|u|=n, u \in T(\omega)} \sum_{i=1}^{N_u} X_u A_{ui} \log \frac{1}{X_u A_{ui}} \middle| \mathscr{F}_n \right]$$

$$= \sum_{|u|=n, u \in T(\omega)} X_u \log \frac{1}{X_u} E \sum_{i=1}^{N} A_i + \sum_{|u|=n, u \in T(\omega)} X_u E \sum_{i=1}^{N} A_i \log \frac{1}{A_i} = Y_n^*.$$

To prove the rest of part (a), we make use of some ideas of Biggins and Kyprianou (1997, Sections 3 and 5) concerning multiplicative martingales. By a theorem of Liu (1998), there is a solution ϕ^* of (E$'$) satisfying

$$\lim_{t \to 0+} \frac{1 - \phi^*(t)}{t |\log t|} = 1.$$

Fix $t > 0$. It is easily seen that the sequence

$$M^{(n)}(t) := \prod_{|u|=n, u \in T(\omega)} \phi^*(X_u t), \quad n > 0,$$

is a martingale associated with $\{\mathscr{F}_n\}$. This martingale is bounded, so has an almost sure and L^1 limit, $M(t)$, satisfying $EM(t) = \phi^*(t)$. Since

$$\lim_{t \to 0+} \frac{-\log \phi^*(t)}{t \log \frac{1}{t}} = \lim_{t \to 0+} \frac{-\log[1 - (1 - \phi^*(t))]}{t \log \frac{1}{t}} = \lim_{t \to 0+} \frac{1 - \phi^*(t)}{t \log \frac{1}{t}} = 1$$

and $\max_{|u|=n, u \in T} X_u \leqslant Y_n \to$ a.s., we see that, for almost all ω and all $\varepsilon > 0$, there is $n(\omega, \varepsilon) \in \mathbb{N}$ such that for all $n > n(\omega, \varepsilon)$, the number

$$-\log M^{(n)}(t) = - \sum_{|u|=n, u \in T(\omega)} \log \phi^*(X_u t)$$

lies between

$$(1 \pm \varepsilon) \sum_{|u|=n, u \in T(\omega)} X_u t \log \frac{1}{X_u t} = (1 \pm \varepsilon)[t Y_n^* + t \log \frac{1}{t} Y_n.$$

Therefore a.s.

$$\lim_{n \to \infty} Y_n^* = -\log M(1) \quad \text{and} \quad \log M(t) = t \log M(1)$$

and $E e^{t \log M(1)} = E e^{\log M(t)} = EM(t) = \phi^*(t)$. Taking $Z^* = -\log M(1)$ ends the proof of part (a).

The proof of part (b) is easy, using the stable transformation due independently to Durrett and Liggett (1983) and Guivarc'h (1990). In fact, applying Theorem 2.0(a) for $\{A_i^\alpha\}$ instead of $\{A_i\}$, we see that $Y_n^{(\alpha)}$ converge a.s. to a non-negative random Z_α with

$$\lim_{t\to 0+} \frac{1-\phi(t)}{t} = 1 \quad \text{where } \phi(t) = E\mathrm{e}^{-tZ_\alpha}.$$

(That is, $EZ_\alpha = 1$.) Since $\phi(t) = E\prod_{i=1}^N \phi(A_i^\alpha t)$, it is easily verified that $\phi_\alpha(t) = \phi(t^\alpha)$ satisfies (E'). To see that ϕ_α is a Laplace transform of a probability distribution on $[0,\infty)$, it suffices to notice that, if X is a non-negative random variable with Laplace transform $\phi_X(t) = \mathrm{e}^{-t^\alpha}$ and independent of Z_α, then ϕ_α is the Laplace transform of $XZ_\alpha^{1/\alpha}$. This gives part (b) of the theorem. $\square$

Note added in proof. This work corresponds to the preprint Liu (1997b). Since the paper was written, A. Rouault has attracted our attention to a paper by J. Neveu [Multiplicative martingales for spatial branching processes, in *Seminar in Stochastic Processes* (ed. E. Cinlar, K.L. Chung and R.K. Getoor), Progress in Probability and Statistics, 1983, 15, 223–242, Birkhäuser, Boston], where a result similar to Theorem 2.5(a) was established for the branching Brownian motion. In a work independent of ours, A.E. Kyprianou [Slow variation and uniqueness of solutions to the functional equation in the branching random walk, J. Appl. Prob. 1998, 35, 795–801] has obtained a variant of Theorem 2.5(a) for the branching random walk.

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References

Athreya, K.B., 1971. A note on a functional equation arising in Galton–Watson branching processes. J. Appl. Probab. 8, 589–598.

Athreya, K.B., Ney, P.E., 1972. Branching Processes. Springer, Berlin.

Barral, J., 1997. Continuté, Moments d'ordres négatifs et Analyse Multifractale de cascades multiplicatives de Mandelbrot. Thèse, Univ. Paris-Sud, Orsay.

Ben Nasr, F., 1987. Mesures aléatoires de Mandelbrot associées à des substitutions. CRAS, Sér. I 304, 255–258.

Biggins, J.D., 1977. Martingale convergence in the branching random walk. J. Appl. Probab. 14, 25–37.

Biggins, J.D., 1979. Growth rates in the branching random walk. Z. Wahrsch. verw. Gebeite 48, 17–34.

Biggins, J.D., 1992. Uniform convergence of martingales in the branching random walk. Ann. Probab. 20, 137–151.

Biggins, J.D., Kyprianou, A.E., 1997. Seneta-Heyde norming in the branching random walk. Ann. Probab. 25, 337–360.

Bingham, N.H., Doney, R.A., 1974. Asymptotic properties of supercritical branching processes I: The Galton-Watson process. Adv. Appl. Probab. 6, 711–731.

Bingham, N.H., Doney, R.A., 1975. Asymptotic properties of supercritical branching processes II: Crump-Mode and Jirina processes. Adv. Appl. Probab. 7, 66–82.

Bingham, N.H., Goldie, C.M., Teugels, J.L., 1987. Regular Variation. Cambridge University Press, Cambridge.

Chauvin, B., Rouault, A., 1996. Boltzmann-Gibbs weights in the branching random walk. IMA Congress on Branching Processes no 84. Lecture Notes in Maths.

Collet, P., Koukiou, F., 1992. Large deviations for multiplicative chaos. Comm. Math. Phys. 147, 329–342.

Dai, C.S., Taylor, S.J., 1992. Defining fractals in a probability space. Preprint.

Doney, R.A., 1972. A limit theorem for a class of supercritical branching processes. J. Appl. Probab. 9, 707–724.

Doney, R.A., 1973. On a functional equation for general branching processes. J. Appl. Probab. 10, 497–508.

Durrett, R., Liggett, T., 1983. Fixed points of the smoothing transformation. Z. Wahrsch. verw. Gebeite 64, 275–301.

Falconer, K.J., 1986. Random fractals. Math. Proc. Cambridge Philos. Soc. 100, 559–582.

Falconer, K.J., 1987. Cut set sums and tree processes. Proc. Amer. Math. Soc. (2) 101, 37–346.

Feller, W., 1971. An Introduction to Probability Theory and its Applications, Vol. II, 2nd Edition. Wiley, New York.

Franchi, J., 1993. Chaos multiplicatif: Un traitement simple et complet de la fonction de partition. Séminaire de probabilités XXIX, Lecture Notes in Mathematics, Springer, Berlin, pp. 194–201.

Grintsevichyus, A.K., 1974. On the continuity of a sum of dependent variables connected with independent walks on lines. Theory Probab. Appl. 19, 163–168.

Grintsevichyus, 1975. One limit distribution for a random walk on the line. Lithunian Math. Trans. 15, 580–589.

Guivarc'h, Y., 1990. Sur une extension de la notion de loi semi-stable. Ann. IHP 26, 261–285.

Harris, T.E., 1948. Branching processes. Ann. Math. Statist. 19, 474–494.

Hawkes, J., 1981. Trees generated by a simple branching process. J. London Math. Soc. 24, 373–384.

Holley, R., Liggett, T., 1981. Generalized poltach and smoothing processes. Z. Wahrsch. verw. Gebeite 55, 165–195.

Holley, R., Waymire, E.C., 1992. Multifractal dimensions and scaling exponents for strongly bounded cascades. Ann. Appl. Probab. 2, 819–845.

Kahane, J.P., 1987. Multiplications aléatoires et dimension de Hausdorff. Ann. IHP, Sup. au no. 2 23, 289–296.

Kahane, J.P., Peyrière, 1976. Sur certaines martingales de Benoit Mandelbrot. Adv. Math. 22, 131–145.

Kesten, H., 1973. Random difference equations and renewal theory for products of random matrices. Acta Math. 131, 207–248.

Kesten, H., Stigum, B.P., 1966. A limit theorem for multidimensional Galton-Watson processes. Ann. Math. Statist. 37, 1211–1223.

Liu, Q.S., 1993. Sur quelques problèmes à propos des processus de branchement, des flots dans les réseaux et des mesures de Hausdorff associées. Thèse, Université Paris 6.

Liu, Q.S., 1996a. The exact Hausdorff dimension of a branching set. Probab. Theory Related Fields 104, 1996.

Liu, Q.S., 1996b. The growth of an entire characteristic function and the tail probabilities of the limit of a tree martingale. In: Chauvin, B., Cohen, S., Rouault, A. (Eds.), Trees. Progress in Probability, Vol. 40. Birkhöuser, Basel, pp. 51–80.

Liu, Q.S., 1997a. Sur une équation fonctionnelle et ses applications: une extension du théorème de Kesten-Stigum concernant des processus de branchement. Adv. Appl. Probab. 29, 1997, 353–373.

Liu, Q.S., 1997b. Self-similar cascades and the branching random walk. Prépublication 97-03, Univ. Rennes 1.

Liu, Q.S., 1998. Fixed points of a generalized smoothing transformation and applications to branching processes. Adv. Appl. Probab. 30, 85–112.

Liu, Q.S., Rouault, A., 1996. On two measures defined on the boundary of a branching tree. In: Athreya, K.B., Jagers, P. (Eds.), Classical and Modern Branching Processes, IMA Volumes in Mathematics and its Applications, Vol. 84. Springer, Berlin, pp. 187–202.

Lyons, R., 1996. A simple path to Biggins' martingale convergence for branching random walk. In: Athreya, K.B., Jagers, P. (Eds.), Classical and Modern Branching Processes, IMA Volumes in Math. and its Appl., Vol. 84, Springer, Berlin, pp. 217–221.

Lyons, R., Pemantle, R., Peres, Y., 1995. Ergodic theory on Galton-Watson trees: Speed of random walk and dimension of harmonic measure. Ergodic Theory Dyn. Systems 15, 593–619.

Mandelbrot, B., 1974. Multiplications aléatoires et distributions invariantes par moyenne pondérée aléatoire, Vol. 278. CRAS, Paris, pp. 289–292, 355–358.

Mauldin, R.A., Williams, S.C., 1986. Random constructions, asymptotic geometric and topological properties. Trans. Amer. Math. Soc. 295, 325–346.

Molchan, G.M., 1996. Scaling exponents and Multifractal Dimensions for independent random cascades. Comm. Math. Phys. 179, 681–702.

Neveu, J., 1986. Arbre et processus de Galton-Watson. Ann. Inst. Henri Poincaré 22, 199–207.

Peyrière, J., 1977. Calcul de dimensions de Hausdorff. Duke Math. J. 44, 591–601.

Rösler, U., 1992. A fixed point theorem for distributions. Stochastic processes and their applications 42, 195–214.

Seneta, E., 1968. On recent theorems concerning the supercritical Galton-Watson process. Ann. Math. Statist. 39, 2098–2102.

Seneta, E., 1969. Functional equations and the Galton-Watson process. Adv. Appl. Probab. 1, 1–42.

Taylor, S.J., 1986. The measure theory of random fractals. Math. Proc. Cambridge Philos. Soc. 100, 383–406.

Taylor, S.J., Tricot, C., 1985. Packing measure, and its evaluation for a brownian path. Trans. Amer. Math. Soc. 288, 679–699.

Waymire, E.C., Williams, S.C., 1995. Multiplicative cascades: dimension spectra and dependence. J. Fourier Anal. Appl., Kahane Special Issue, 589–609.